Liquidation Mechanism Design and Risk Analysis in Decentralized Perpetual Exchanges

A Comparative Study of GMX, dYdX v4, and Hyperliquid

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1. Abstract

Decentralized perpetual exchanges (perp DEXes) now process tens of billions of USD in daily notional volume. Their liquidation engine is the single most safety-critical component: it decides who pays for a losing trade, how fast bad debt is closed, and ultimately who absorbs losses when markets gap. This report compares three architecturally distinct platforms —**GMX** (pool-counterparty AMM-style perps), **dYdX v4** (validator-run on-chain CLOB), and **Hyperliquid** (HyperCore L1 CLOB with a protocol-owned backstop vault) —across leverage design, liquidation logic, oracle integrity, and extreme-volatility defenses. We then

dive into GMX's liquidation flow, quantify the GM-pool LP impact with a worked example, and close with a risk taxonomy and forward-looking design directions illustrated by real incidents including the **Hyperliquid JELLY squeeze of March 2025**.

2. Background: Anatomy of a Perp Liquidation

A perpetual contract is a derivative with no expiry; it tracks an index price through a **funding rate** that periodically transfers value between longs and shorts. Each position posts **initial margin (IM)** as collateral; the position becomes liquidatable when account equity falls below **maintenance margin (MM)**.

Key concepts used throughout this report:

- **Mark price**: the price used for PnL and liquidation checks, distinct from the last trade price.
- **Liquidator / keeper**: the actor that submits the closing transaction in exchange for a reward.
- **Insurance fund (IF)**: pooled capital that absorbs shortfalls when the liquidation price is worse than bankruptcy.
- **Auto-deleveraging (ADL)**: a last-resort socialization mechanism that force-closes profitable positions to clear bad debt.

Designers face a fundamental three-way trade-off: **user experience** (high leverage, low fees, partial liquidations), **system safety** (conservative MM, large IF, fast oracles), and **decentralization** (permissionless keepers, no admin pause). The three platforms studied here sit at different points on this triangle.

3. Platform Architectures Side-by-Side

Dimension	GMX v2 `[GMX-OV]`	dYdX v4 `[DYDX-PORT]`	Hyperliquid `[HL-LIQ]`
Venue	Arbitrum / Avalanche L2	Cosmos SDK app-chain	HyperCore L1 (custom BFT chain)
Matching	No order book —oracle-priced AMM-style fills against GM pool	On-chain central limit order book (CLOB) run by validators	On-chain CLOB run by validators
Counterparty	GM / GLV liquidity pool (LPs)	The other side of the book + insurance fund	The other side of the book + HLP backstop vault
Custody	Non-custodial smart contracts	Funds held in chain modules	Funds held in HyperCore L1 state
Settlement unit	USD-denominated; collateral in pool tokens (e.g. WETH + USDC)	USDC	USDC
Liquidator role	Permissionless keeper bots	Any full-node operator running the liquidation daemon	Any market participant (market liq) + HLP vault (backstop)
Oracle	Chainlink Data Streams (signed pull-based) `[GMX-ORD]`	Validator vote-extension aggregation `[DYDX-ORC]`	Validator-weighted median of CEX spot prices, 3-second cadence `[HL-ORC]`

The architectures encode three philosophies. GMX shifts counterparty risk onto a passive LP pool and accepts oracle dependence as the price of zero order-book infrastructure. dYdX rebuilds a CEX-style CLOB inside a sovereign chain, treating the validator set itself as the matching engine and price source. Hyperliquid blends both: a CLOB for normal flow, plus an HLP vault that acts as a community-funded market maker and insurance fund in one.

4. Leverage and Margin Design

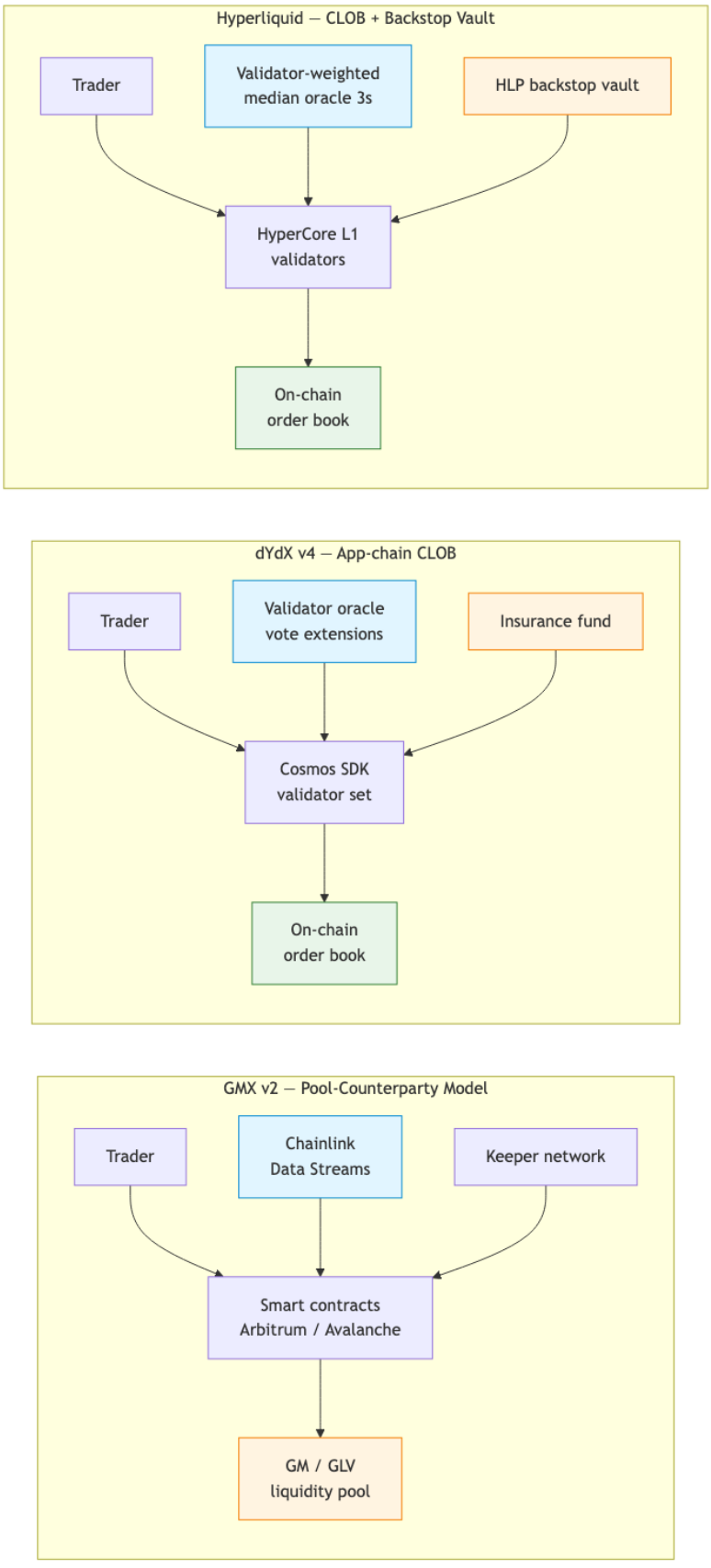


Figure 1: Three-platform architecture comparison

Item	GMX [GMX-LIQ]	dYdX v4 [DYDX-MGN]	Hyperliquid [HL-TIER]
Max leverage	up to 100× crypto; 25–40× TradFi off-hours	1–50× depending on market liquidity tier	3–40× per asset (BTC 40× → 20× above $150M OI; ETH 25× → 15×)
Margin mode	pool-collateralized; no formal cross/iso	cross (default) + isolated subaccounts	cross (default) + isolated; tiered MM table
Initial margin	market-dependent floor	IM = $\lvert S \cdot P \rvert \cdot$ IMF, IMF 2–100 %	IM = notional/leverage
Maint. margin	0.25–1 % of notional (asset floor)	fixed MMF per market	MMR = IMR/2 at max lev.; widens by tier
Liq. rule	`collateral < min_threshold`	`account_value < total_MMR`	`equity < MM_required`

Hyperliquid's tiered structure is the most aggressive on UX (40× BTC is rare in DeFi) but the protocol claws safety back by reducing leverage as a single account's open interest grows —a structural defense against whale-driven cascades. dYdX's MMF is **fixed per market** and **scales with open interest caps**, providing predictable margin requirements at the cost of less granularity. GMX's "minimum collateral threshold"model is the simplest from a user mental-model perspective but couples liquidation tightly to oracle freshness.

5. Liquidation Logic Comparison

5.1 Trigger and executor

- **GMX [GMX-LIQ]** —A permissionless keeper network monitors on-chain position state. When `collateral < min_collateral_threshold` (0.25 %–1 % of notional, market-dependent), any keeper may submit a liquidation transaction carrying a signed Chainlink Data Streams report. Execution is two-phase: the oracle price is committed in block N and the position closes in block $N+1$, reducing keeper MEV opportunities [GMX-ORD].
- **dYdX v4 [DYDX-LIQ]** —The chain itself detects accounts below MMR each block. A liquidation order is **automatically generated by the protocol** and placed on the open CLOB. Any node running the liquidation daemon competes to fill it.
- **Hyperliquid [HL-LIQ]** —A three-tier waterfall:
  1. **Market liquidation** —the first 20 % of the position is sent to the book as a market order, with a 30-second cooldown before the rest;
  2. **HLP backstop** —if fewer than two-thirds of required MM can be filled, the remaining position is transferred wholesale to the HLP vault;
  3. **ADL** —if HLP cannot absorb the loss, profitable counter-positions are force-closed at mark price.

5.2 Liquidation price and penalty

- **GMX**: closes at the worse side of the oracle spread (longs at `maxPrice`, shorts at `minPrice`). Liquidation fee is **0.20 %** for standard markets and **0.45 %** for synthetic / high-volatility markets, deducted from the position notional [GMX-FEE].
- **dYdX**: closes at a *fillable price* P_{fill} derived from the oracle by widening the spread toward bankruptcy as the account deteriorates:

$$P_{\text{fill}} \;=\; P \cdot \Big(1 \;-\; s \cdot \text{SMMR} \cdot \text{MMF} \cdot \text{BA} \cdot (1 - Q) \Big)$$

  with the following variable convention [DYDX-LIQ]:

  – $s = +1$ when liquidating a **long** (sell side, fill below oracle) and $s = -1$ for a **short** (buy side, fill above oracle), guaranteeing the fill is always adverse to the liquidatee;
  – SMMR is the protocol-wide *Spread-to-Maintenance-Margin Ratio* (currently 1.0–1.5), which caps how much of the maintenance band can be paid out as liquidation discount;

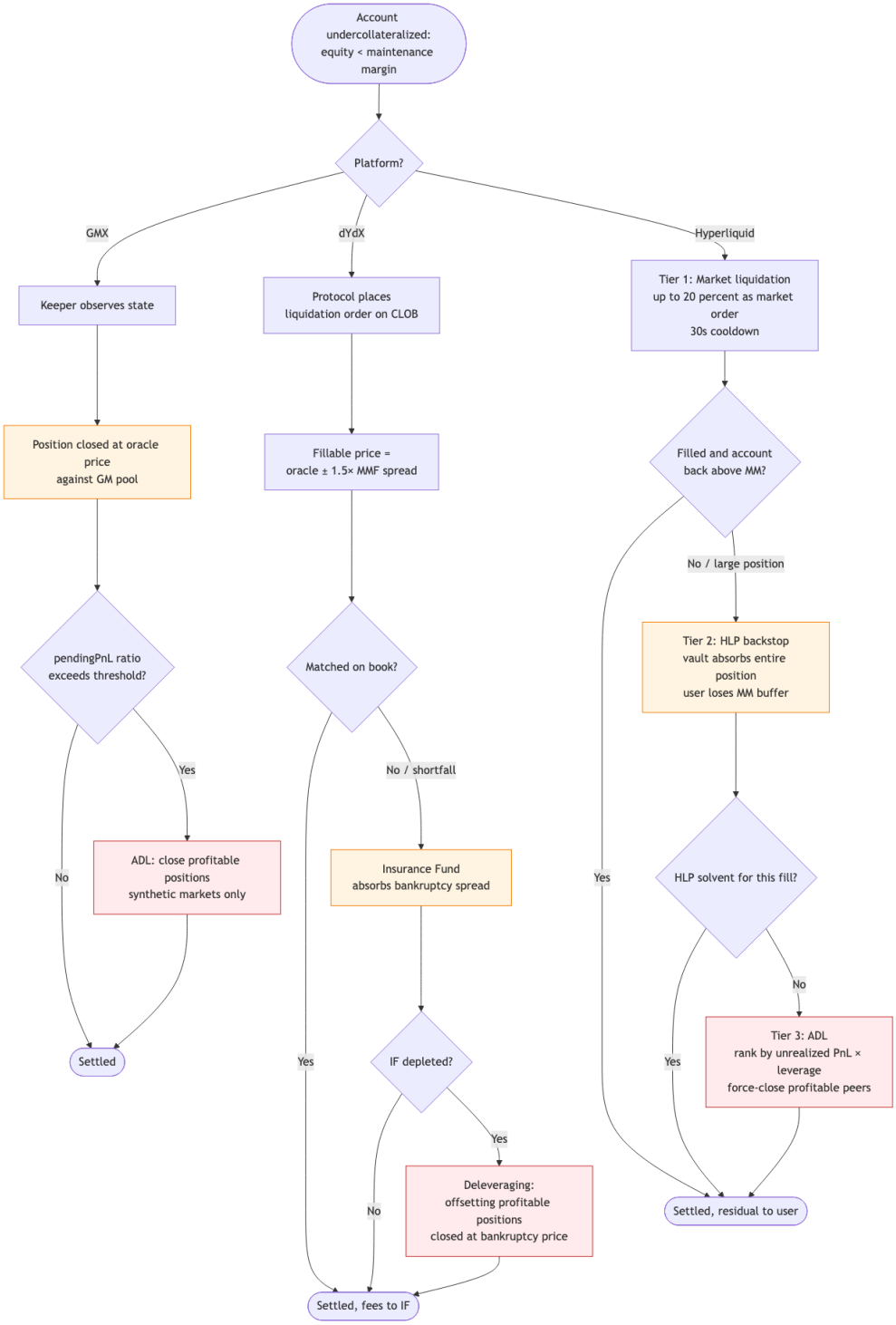


Figure 2: Liquidation waterfall across three platforms

– MMF is the per-market maintenance-margin fraction (e.g. 0.03 for BTC-USD);
– BA $\in [0,1]$ is the **bankruptcy-adjustment** factor: it linearly shrinks the spread to 0 as P_{fill} approaches the bankruptcy price, preventing the fill from being *worse* than the account's zero-equity price;
– $Q = \text{equity}/\text{MMR} \in [0,1)$ measures how deep the account has fallen into the maintenance region ($Q \to 0$ at bankruptcy).

The product $\text{SMMR} \cdot \text{MMF} \cdot (1-Q)$ is itself capped, enforcing the headline $\leq 1.5\%$ **of account value** liquidation-penalty fee routed to the insurance fund.

- **Hyperliquid**: the on-chain liquidation price is

$$P_{\text{liq}} \;=\; P_{\text{mark}} \;-\; s \cdot \frac{\text{margin}_{\text{avail}}}{|S|} \cdot \frac{1}{1 - l/l_{\text{MM}}}$$

with $s = +1$ for longs and $s = -1$ for shorts; l is the account's current leverage and l_{MM} the tier-specific maximum maintenance leverage `[HL-MGN]`. There is **no explicit liquidation fee**; once the position falls into HLP, the trader simply forfeits their remaining margin buffer.

Edge-case behaviour. As $l \to l_{\text{MM}}$ from below, the factor $1/(1 - l/l_{\text{MM}})$ diverges to $+\infty$, which would imply an arbitrarily large mark-price distance to liquidation. Hyperliquid handles this discontinuity in two layers. (i) The protocol re-evaluates the tier MM table whenever the position OI crosses a tier boundary, so a position that approaches its tier ceiling is **forcibly bumped to the next tier**'s lower l_{MM}, deflating the divergent term back to a finite value `[HL-TIER]`. (ii) The liquidation check itself is on equity $< \text{MM}_{\text{req}}$, not on P_{liq}, so the singularity is mathematical rather than operational; the formula is published for traders to visualise their buffer, while the on-chain engine never divides by a quantity smaller than the discrete tier step. Liquidation-bot authors should therefore **clip** l/l_{MM} at $1-\epsilon$ (with $\epsilon \approx 10^{-3}$) and **fallback** to the equity-vs-MM check at the boundary, which is the same invariant the chain itself enforces. This is the canonical edge-case guard for any keeper implementation.

5.3 Liquidator incentives

Platform	Incentive	MEV exposure
GMX	keeper subsidy from protocol fees (~10 % of v2 fees; 1.2 % to Chainlink) `[GMX-TOK]`	low —two-phase oracle commitment
dYdX v4	implicit profit from filling at the fillable-price discount; competitive among full nodes	medium —orders visible in mempool / CLOB
Hyperliquid	market liq is open competition (price improvement = profit); HLP earns the backstop spread `[HL-LIQ]`	low —validators set mark price every 3 s

6. Oracle Design and Manipulation Defenses

Property	GMX	dYdX v4	Hyperliquid
Primary source	Chainlink Data Streams (signed pull)	Validator side-cars query multiple CEXes	Validator-weighted median of CEX spot
Sources aggregated	Chainlink internal feed network	Binance, Coinbase, Bybit, Bitfinex, Kraken, etc.	Binance 3×, OKX 2×, Bybit 2×, Kraken, Kucoin, Gate, MEXC `[HL-ORC]`
Update cadence	Sub-second (pull on execution)	Each block (~1 s) via vote extensions	Every **3 seconds**
Aggregation	Provider-internal	Weighted consensus across validator votes	Stake-weighted median per asset

Property	GMX	dYdX v4	Hyperliquid
Bid / ask spread	`minPrice` / `maxPrice` reported	Single index price	Single mark price
Manipulation defense	Pull-based signed reports; two-phase commit	Validator quorum; governance over source list	OI caps + 1 % resting-order distance cap + HLP exemption `[HL-RISK]`

The defining trade-off is **who you trust**: GMX trusts the Chainlink ecosystem (auditable, off-chain, but a single provider dependency); dYdX trusts its validator set to honestly query and report exchange data; Hyperliquid trusts its validators *and* uses CEX liquidity weights as a sybil-resistance heuristic. None of the three rely on a thin internal AMM as oracle —a deliberate response to the wave of 2021–2022 spot-AMM oracle exploits.

7. Extreme-Volatility Defenses

Mechanism	GMX	dYdX v4	Hyperliquid
Circuit breakers	Price-impact caps (0.5–10 %); off-hours leverage cut `[GMX-LIQ]`	None explicit	OI caps + 1 % resting-order distance cap (HLP exempt) `[HL-RISK]`
Delay auction	Two-phase keeper execution (~1 block)	None	3-s oracle granularity + 30-s liq cooldown
Backstop liquidity	GM / GLV pool (capped by `MAX_PNL_FACTOR`)	Insurance fund (community-funded)	HLP vault (community-funded)
ADL trigger	`pendingPnL / poolValue > MAX_PNL_FACTOR_FOR_ADL` (synthetics)	Account equity < 0 (bad debt)	HLP cannot absorb the entire underwater position
ADL victim selection	Most-profitable open positions in that market	Ranked by unrealised-profit and position-size tier at the bankruptcy price `[DYDX-LIQ]`	Ranked by $\frac{P_{\text{mark}}}{P_{\text{entry}}} \cdot \frac{\text{notional}}{\text{equity}}$ `[HL-ADL]`

Three different answers to the same problem. GMX makes its own LPs the backstop, ring-fencing pool exposure via `MAX_PNL_FACTOR`. dYdX raises a community-governed insurance fund and only socializes losses after it is exhausted. Hyperliquid packages backstop liquidity and LP returns into one product (HLP) — a clever incentive alignment, but one that concentrates risk in a single vault, as the JELLY incident later illustrated.

8. Funding Rate Mechanism

Property	GMX `[GMX-FEE]`	dYdX v4 `[DYDX-FUND]`	Hyperliquid `[HL-FUND]`
Formula type	Adaptive (long/short OI imbalance, three-zone threshold)	Premium index + interest rate component	Premium index + interest rate (0.01 % floor)
Sampling	Continuous per-second accrual	1-minute premium samples	5-second premium samples
Settlement	On position modification / closure (checkpoint)	Hourly	Hourly (1/8 of 8-hour rate)

Property	GMX [GMX-FEE]	dYdX v4 [DYDX-FUND]	Hyperliquid [HL-FUND]
Cap	15–300 % APR, governed by keeper from reference CEX funding	$600\% \cdot (\text{IMF} - \text{MMF})$ per 8 h	4 % per hour
Paid in	Pool token PnL	USDC	USDC, computed against **oracle price** (not mark)

9. GMX Deep-Dive: Liquidation Flow and LP Impact

9.1 End-to-end walkthrough

1. **Monitoring** —A permissionless keeper network indexes open positions and continually evaluates the maintenance-collateral condition using on-chain state plus the latest Chainlink Data Streams report [GMX-LIQ].
2. **Oracle commit** —When a position falls below threshold, the keeper assembles a transaction containing the signed oracle report (with `minPrice` and `maxPrice`) and submits it in block *N*.
3. **Execution** —In block *N+1*, the protocol closes the position at the *adverse* side of the oracle spread (longs at `maxPrice`, shorts at `minPrice`) to protect pool LPs from oracle latency [GMX-ORD].
4. **Fee deduction** —A liquidation fee of **0.20 %** (standard markets such as ETH-USD) or **0.45 %** (synthetic markets such as DOGE-USD) is deducted from position notional [GMX-FEE].
5. **Fee split** —Of the collected fee plus the trader's lost margin and borrow fees, **63 % flows to the GM pool LPs** and **37 % to the protocol treasury / GMX stakers** [GMX-LP].
6. **Pool settlement** —The pool realizes the counterparty PnL: it gains the trader's loss in the appropriate pool token (e.g. WETH for a long ETH liquidation, USDC for a short).
7. **ADL check** —If the realized + pending PnL of the surviving traders against the pool exceeds the market's `MAX_PNL_FACTOR_FOR_ADL`, GMX auto-deleverages the most profitable positions to restore pool solvency. This is principally a defense for **synthetic markets**, where the pool's collateral token differs from the indexed asset.
8. **Residual return** —Any collateral remaining after fees and PnL is returned to the trader.

9.2 GLP (v1) → GM (v2) evolution

Aspect	GLP (v1)	GM / GLV (v2)
Pool composition	Single index basket (ETH/WBTC/USDC/...)	Per-market pair (e.g. ETH-USDC), with isolated fully-backed pools
Risk surface	All positions hit one shared pool	Per-market isolation limits contagion
LP exposure	Aggregate basket VaR	Targeted, hedgeable per-asset exposure
Oracle	Chainlink price feeds (push)	Chainlink Data Streams (pull, signed)
ADL	Limited	Configurable `MAX_PNL_FACTOR_FOR_ADL` per market [GMX-LP]
Lesson learned	2022 AVAX manipulation (single-pool contagion)	Per-market isolation + adaptive funding contain similar attacks

9.3 Worked LP-impact example

Consider a 10× long of $1,000,000$ notional on ETH-USD (pool: WETH-USDC), entered at \$3,000 with \$100,000 collateral. ETH drops to \$2,730 (9 % move). The trader's PnL is −\$90,000, leaving roughly \$10,000 collateral —below the 1 % maintenance floor. The keeper triggers a liquidation:

- Liquidation fee: $0.20\% \times \$1{,}000{,}000 = \$2{,}000$
- Loss absorbed by pool: +\$90,000 (in WETH terms, since long lost ETH exposure value)

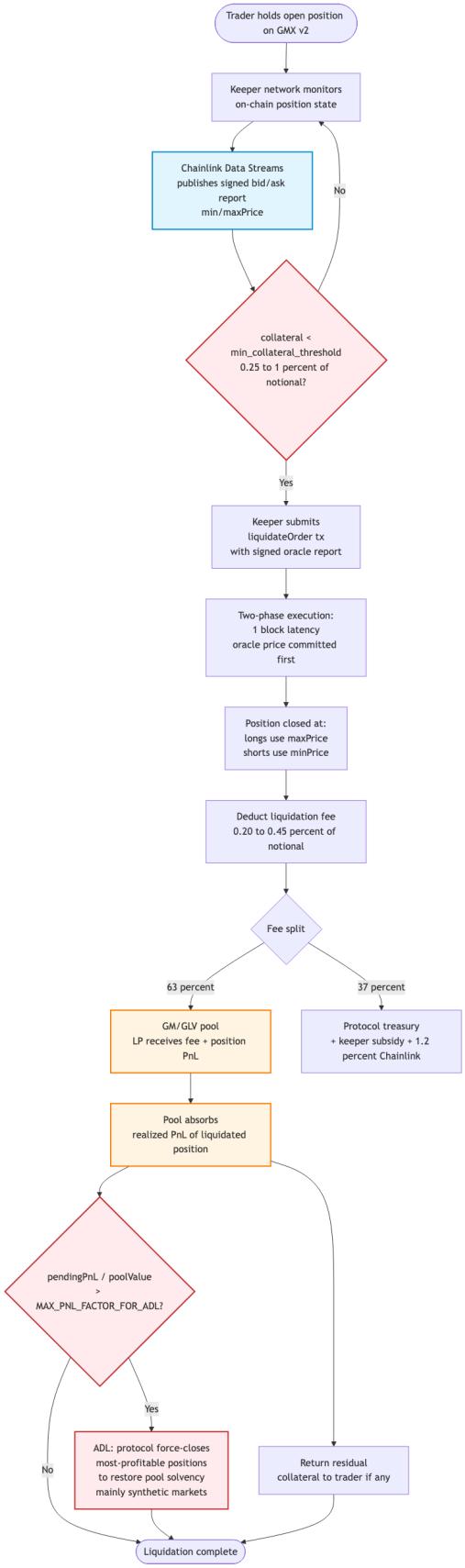


Figure 3: GMX v2 liquidation flow

- Fee split: $0.63 \times \$2{,}000 = \$1{,}260$ to pool LPs, \$740 to protocol
- Residual to trader: $\$10{,}000 - \$2{,}000 \approx \$8{,}000$

The GM pool's NAV rises by approximately \$91,260 from this single liquidation. Symmetrically, if the trade had succeeded, the same magnitude would have been paid *out* of the pool —bounded by `MAX_PNL_FACTOR`, beyond which ADL would have triggered. **The GM pool is therefore a long-volatility counterparty to perp traders**: it profits from liquidations and loses from sustained directional moves that traders capture, very similar in PnL profile to a passive option-writing strategy.

10. Historical Incidents and What They Taught

Date	Event	Root mechanism	Lesson absorbed by modern designs
12 Mar 2020	"Black Thursday" `[EXT-312]`	MakerDAO 0-bid liquidations; Ethereum gas congestion	Need for backstop bidders; minimum bid floors
May 2022	LUNA / UST collapse `[EXT-LUNA]`	Reflexive death spiral in collateral asset	Oracle robustness against collapsing assets; OI / asset caps
Sep 2022	GMX v1 AVAX manipulation `[EXT-GMX1]`	Thin spot-market price + GLP single-pool exposure	Per-market pool isolation (v2); adaptive max leverage caps
Nov 2022	FTX implosion `[EXT-FTX]`	Centralized custody + commingled customer funds	Validation of non-custodial perp DEX thesis
Mar 2025	**Hyperliquid JELLY squeeze** `[EXT-JELLY]`	Low-cap token, attacker self-traded into HLP; HLP took on toxic inventory it could not exit	Asset-onboarding due diligence; tighter OI caps for new listings; HLP delisting governance

The **JELLY incident** in particular spotlights the structural cost of Hyperliquid's HLP design. The vault performed exactly as designed when it accepted a backstop liquidation, but the underlying asset's spot market was illiquid enough that HLP could not unwind without a large loss. Hyperliquid's response —delisting the asset and reimbursing affected users —was effective but required *governance intervention*, partly negating the appeal of an automated backstop. The systemic lesson is that any insurance-fund-like primitive must include **asset onboarding controls** and **per-asset risk budgets**, not just aggregate solvency tracking.

10.1 Worked toxic-inventory transfer: a stylised JELLY model

To balance the GMX worked example in §9.3, we reconstruct a simplified mathematical model of how the JELLY attacker pushed toxic inventory into HLP. Let:

- L = aggregate USDC liquidity quoted within ± 5 % of mid in the JELLY orderbook $\approx \$0.3\,\text{M}$ at the time of attack;
- V = HLP's available USDC backstop budget $\approx \$240\,\text{M}$ `[EXT-JELLY]`;
- l = leverage used by the attacker on the seed short $\approx 20\times$;
- c = attacker's seed collateral $\approx \$3.5\,\text{M}$, giving notional short $N = l\,c \approx \$70\,\text{M}$;
- p_0, p_1 = mark price before and after the attacker's spot ramp; the actual move was $\Delta p/p_0 \approx +4\times$ (i.e. $p_1 = 4p_0$) over minutes.

Stage 1 —seed. From a *self-controlled* address A_1, the attacker opens a high-leverage short of size N on JELLY-PERP. Because $N \gg L$, no honest market maker can hedge it; this is the toxic inventory.

Stage 2 —ramp. From other addresses $A_2, A_3, \ldots$, the attacker buys spot JELLY on the thin DEX market, pushing mark to p_1. The unrealised loss on A_1 is

$$\text{PnL}_{A_1}(p_1) \;=\; -N \cdot \frac{p_1 - p_0}{p_0} \;\approx\; -\$210\,\text{M}.$$

Account equity collapses: $c + \mathrm{PnL}_{A_1} \ll 0$, well past the maintenance band.

Stage 3 —forced HLP transfer. Per §5.1, the liquidation waterfall first tries to market-out 20 % into the book. The book absorbs at most $L \ll 0.2N$, so the post-cooldown remainder —essentially the full N —falls to the HLP backstop. The chain now records HLP as the counterparty to a \$70 M short at the inflated p_1.

Stage 4 —exit asymmetry and the dual-account profit channel. HLP's mark-to-market loss on this inherited position is bounded by *its own* solvency, V. Crucially, the liquidation engine transfers the toxic short to HLP **before** any Tier-3 ADL is invoked, which opens a *liquidity time-lag*: while HLP is still trying to unwind into the same thin orderbook, the attacker's profit-side addresses $(A_2, A_3, ...)$ are already sitting on the realised spot or long-perp gains generated during the ramp. They can withdraw those gains as clean USDC (or simply close into the now-thicker bid that HLP itself provides) in the seconds-to-minutes window before ADL would have force-closed them. The sacrificial A_1 is bankrupt by design, but A_1's loss is *capped at* c; HLP absorbs the rest. The attacker's expected extraction is therefore

$$\mathbb{E}[\text{extraction}] \;\approx\; \min\Big(N \cdot \frac{p_1 - p_0}{p_0},\; V\Big) \;-\; c,$$

i.e. up to $\min(\$210\,\mathrm{M}, \$240\,\mathrm{M}) - \$3.5\,\mathrm{M} \approx \$206\,\mathrm{M}$ of socialised loss before governance intervention. The asymmetry is fundamental: the protocol gives HLP a **single-sided** toxic inventory while letting the attacker exit on the **opposite** side at the same elevated mark.

Why the attack worked: the microstructure inequality. The vulnerable condition is

$$\begin{cases} l \cdot c \;>\; L_{\text{exit}} & \text{(orderbook cannot absorb the seed)} \\ l \cdot c \;\lesssim\; V & \text{(but HLP is rich enough to swallow it)} \end{cases}$$

that is, the seed notional must exceed available exit liquidity L_{exit} (so HLP cannot unwind) yet stay within the backstop budget V (so the position is fully absorbed rather than rejected). For mature assets such as BTC, $L_{\text{exit}} \gg V$ and the attack is impossible. For new, thin listings the inequality flips and HLP becomes the dump destination.

Mitigations that close the inequality. Tighter per-asset OI caps lower the LHS; orderbook-depth-aware HLP admission lowers the RHS budget allocated to the asset; and a *delisting fast-path* (governance or automated, e.g. on $\Delta P/P > k\sigma$) caps Stage 4 by terminating the contract before V is consumed. Hyperliquid post-incident has deployed variants of all three.

11. Key Risks and Future Directions

11.1 Open risks (common to all three)

1. **Oracle compromise** —GMX is exposed to Chainlink Data Streams operator integrity; dYdX and Hyperliquid to validator collusion. A coordinated price push during a thin-liquidity window can trigger cascading liquidations before reversion `[HL-RISK]`.
2. **MEV / front-running of liquidations** —although all three designs reduce it, profitable keepers still incentivize sophisticated bots, concentrating extraction.
3. **ADL fairness** —being force-closed while in profit is the worst UX in perp DEXes. None of the three platforms compensates ADL victims beyond closing at mark.
4. **Smart-contract / chain-halt risk** —dYdX and Hyperliquid run sovereign chains; a validator outage halts both trading *and* liquidation, effectively freezing risk.
5. **LP / HLP impermanent risk** —GM and HLP investors hold short-vol exposure that can decay quickly in a sustained trend (e.g., HLP losses during JELLY).
6. **Bridged collateral and stablecoin depeg** —USDC depegs (Mar 2023) would simultaneously impair all three platforms.
7. **Cross-margin correlation risk** —dYdX and Hyperliquid default to cross-margin (§4), so a single account's equity is the sum over n positions, $E = c + \sum_{i=1}^{n} S_i\,(P_i - P_i^{\text{entry}})$, while the maintenance requirement is computed *per-position* with $\mathrm{MM}_{\text{req}} = \sum_i |S_i|\,P_i\,\mathrm{MMF}_i$. The variance of equity, however,

is $\mathrm{Var}(E) = \sum_{i,j} S_i S_j \sigma_i \sigma_j \rho_{ij}$ with pairwise correlations ρ_{ij}. Because the maintenance sum **ignores** the off-diagonal terms, an account that is long ten correlated alt-coins is treated as ten independent risks even though their joint $\rho_{ij} \to 1$ in a crash. Empirically alt-major correlations spike from ~ 0.5 to > 0.9 during sell-offs (e.g., 12 Mar 2020, May 2022), producing cascading cross-margin liquidations that the per-position MM never priced in. The TradFi fix —a haircut matrix H with $\mathrm{MM}^{\mathrm{corr}}_{\mathrm{req}} = \sqrt{\mathbf{w}^\top H \mathbf{w}}$ and $H_{ij} = \rho_{ij}\,\mathrm{MMF}_i\,\mathrm{MMF}_j$ —is not directly portable on-chain: maintaining and re-pricing an $n \times n$ matrix every block is $O(n^2)$ per account and $O(n^3)$ amortised across the validator set, which on a CLOB chain processing 10^4 orders/s is prohibitively expensive (HyperCore L1 currently budgets a few microseconds per margin check). The pragmatic engineering compromise is the **correlation-group / asset-class haircut** approach used by legacy FTX and CME SPAN: bucket assets into a handful of risk classes ($K \ll n$, e.g. {BTC, ETH, majors, mid-caps, long-tail}), apply a step-wise add-on to the dominant net exposure within each class, and assume $\rho = 1$ inside the class. This collapses the per-account cost back to $O(n + K)$ while still capturing the *first-order* cascade risk that the diagonal-only model misses.

8. **Funding-induced liquidation cascades** —the per-second / per-hour funding accruals analysed in §8 directly bite into the same `margin_avail` that drives the Hyperliquid liquidation price in §5.2. In a short squeeze on a thin asset (the JELLY pattern of §10.1), the imbalance pushes the funding rate to its cap —**4 % per hour** on Hyperliquid, equivalent to a $\sim 960\%$ APR —so a short with one-hour margin runway loses an additional $0.04 \cdot N$ to funding *before* mark moves. Concretely, the maintenance buffer evolves as
$$\frac{d\,\mathrm{margin}_{\mathrm{avail}}}{dt} \;=\; -f(t) \cdot N \;-\; S \cdot \frac{dP_{\mathrm{mark}}}{dt},$$
with $|f(t)| \le 0.04$/h. The first term alone can flip a 5% safety margin to liquidatable in ≈ 1.25 hours of capped funding, *independently* of price action, so the liquidation cascade can self-sustain even on a temporarily flat mark. None of the three platforms currently dampens the funding term as accounts approach MM (a `funding\_taper` would be a natural mitigation, mirroring the bankruptcy-adjustment factor BA in §5.2).

11.2 Forward-looking design ideas

As a systems-engineering exercise, each idea below is paired with the latency / throughput / liveness trade-off it introduces.

- **ZK-proven oracles** (e.g., Pyth Lazer with succinct proofs, or zkTLS-attested CEX prices) reduce trust in any single signer. *Trade-off:* current prover latencies for Groth16 / Plonky2-class circuits sit at ~ 200 ms–2 s per attestation —acceptable for GMX's pull-on-execute model, but already comparable to Hyperliquid's 3-second oracle cadence and roughly an order of magnitude above per-block CLOB matching latency. A naive ZK feed would therefore *increase* effective oracle staleness; production designs will likely need recursive proof aggregation off the hot path and signed inclusion proofs on the hot path.
- **Intent-based liquidation auctions** —instead of first-come keepers, route liquidations through batch auctions (à la CoW Swap) so that liquidator profit accrues to the user. *Trade-off:* the auction window τ directly extends the time during which the underwater position remains open. In a regime where mark moves at velocity v (USD/s), the expected additional bad debt scales as $\Delta D \sim |S| \cdot v \cdot \tau$. For BTC, v can reach 10^3–10^4 USD/s in a flash crash, so even $\tau = 500$ ms can accumulate non-trivial debt. Auctions are therefore a UX win in calm markets but a tail-risk amplifier in crashes; a hybrid (immediate keeper fill in stressed regimes detected by realised vol, auction in calm regimes) is the obvious compromise.
- **Hybrid CLOB + RFQ for liquidation flow** —let market makers quote firm prices for liquidated inventory before it hits public order books, reducing impact. *Trade-off:* RFQ latency (round-trip to MM + response window) lengthens execution; if no MM responds (the regime when one most needs them), the system must fall back to CLOB instantly, doubling the engineering surface.
- **Per-asset insurance tranches** —replace monolithic IF / HLP with risk-isolated vaults priced by market makers, preventing one toxic asset (JELLY-style) from draining the whole backstop. *Trade-off:*

fragmentation reduces capital efficiency; vaults backing thin assets demand higher yields, raising trader funding costs on exactly those markets.

- **Formal verification of margin invariants** —proving that equity − MM ≥ 0 is preserved across all reachable states would eliminate an entire bug class. *Trade-off:* development velocity collapses; protocol upgrades require re-proving the invariant, which has historically been a multi-month effort for non-trivial state machines.
- **Programmable circuit breakers** —auto-widen MM and pause new leverage on detected volatility spikes, similar to NYSE LULD bands but on-chain and parameterized per asset. *Trade-off:* false positives on whipsaw price prints can freeze legitimate hedging exactly when it is most needed; calibration of the volatility detector itself becomes a governance risk.

12. Conclusion

The three platforms encode three coherent worldviews about how a perpetual exchange should be safe:

- **GMX** says safety = isolating risk on willing LPs, capping their exposure with `MAX_PNL_FACTOR`, and trusting Chainlink for prices. UX is good for traders, demanding for LPs.
- **dYdX v4** says safety = a faithful reproduction of CEX mechanics inside an open sovereign chain, with a transparent insurance fund as the shock absorber. UX is most familiar to professional traders.
- **Hyperliquid** says safety = aggressive leverage backed by an automated community vault, accepting that the vault will occasionally take losses (JELLY) in exchange for unmatched UX. UX is best in class; safety leans on governance to fix edge cases.

None of the designs is strictly dominant. As perpetual DEXes converge toward centralized-exchange functionality, the next round of innovation will likely re-mix these primitives —oracles from one, insurance from another, matching from a third —into hybrid architectures that try to break, rather than navigate, the UX-vs-safety trade-off.

Appendix A. Symbol Glossary

Symbol	Meaning
S	Signed position size (positive = long)
P	Oracle / mark price
IM	Initial margin
MM	Maintenance margin
IMF	Initial margin fraction
MMF	Maintenance margin fraction
OI	Open interest
IF	Insurance fund
ADL	Auto-deleveraging
HLP	Hyperliquidity Provider vault
GM	GMX v2 per-market pool
GLV	GMX v2 vault aggregating GM pools

Appendix B. Bibliography

Citation keys are resolved in refs.md. Primary sources are the official documentation of each platform; secondary sources for incidents are platform post-mortems and contemporaneous press coverage referenced in §10.